1. **To learn and execute basic Linux command-line operations for file management, directory navigation, file viewing, and obtaining command help through the terminal.**

Objective

To introduce students to the Linux Command-Line Interface (CLI), establishing foundational skills for navigating the filesystem hierarchy, performing file and directory management, inspecting textual data, and utilizing built-in documentation to independently discover command parameters.

Learning Outcomes

By the end of this lab, students will be able to:

- **Navigate** the Linux directory tree using absolute and relative pathnames with pwd, cd, and ls.
- **Perform** standard file and directory management tasks, including creation (touch, mkdir), copying (cp), moving/renaming (mv), and removal (rm, rmdir).
- **Inspect and analyze** text file contents using whole-file output (cat), paginated viewing (less), and head/tail viewing (head, tail).
- **Retrieve** command documentation and syntax assistance directly from the terminal using man, whatis, info, and --help flags.

Procedure:

1.1. Environment Navigation & Inspection: Directory Traversal.

Verify your current location in the filesystem hierarchy and explore directory contents using various flags.

Bash

Print absolute path of current working directory

pwd

List contents in current directory (standard view)

ls

List detailed information including permissions, file sizes, and modification dates

ls -l

List all files including hidden files (starting with .)

ls -la

Change directory to the system temporary directory, then return to home

cd /tmp

pwd

cd ~

Linux / macOS (bash, zsh)

Goal	Command
Go to your home directory	cd (no argument), cd ~, or cd $HOME
Go to another user's home	cd ~username
Go to the root directory (/)	cd /
Go back to the previous directory	cd -
Go up one level	cd ..

$ pwd
/home/user/projects/app
$ cd ..
$ pwd
/home/user/projects # parent of app
$ cd -
/home/user/projects/app # back to where you were before

2.2. Directory & File Creation: Creation & Structure.

Construct a structured directory hierarchy and create sample text files.

```
# Create a project folder structure

mkdir -p ~/cli_lab/notes ~/cli_lab/data


# Change into the project directory

cd ~/cli_lab


# Create empty text files using 'touch'

touch notes/day1.txt notes/day2.txt data/dataset1.csv


# Verify structure creation

ls -R
```

Note: ls -R lists the contents of the current directory **recursively**: it shows every file and folder in it, then goes into each subdirectory and lists those too.

3.3. File Copying, Moving, and Renaming: File Manipulation.

Manipulate files across directories and perform renaming operations.

```
# Add text content to a file using redirection

echo "Linux CLI Fundamentals - Day 1 Notes" > notes/day1.txt

echo "Linux CLI Fundamentals - Day 2 Notes" > notes/day2.txt


# Copy a file to create a backup copy

cp notes/day1.txt notes/day1_backup.txt
```

Move a file to another directory

mv notes/day1_backup.txt data/

Rename a file using 'mv'

mv notes/day2.txt notes/day2_archived.txt

Verify operations

ls -l notes data

4.4. Viewing File Contents: File Content Inspection.

Inspect text file contents using different utilities tailored for short and long files.

Display entire file contents directly to standard output

cat notes/day1.txt

Concatenate and view multiple files with line numbers

cat -n notes/day1.txt notes/day2_archived.txt

View a long system file interactively (Press 'q' to quit less)

less /etc/services

View the first 5 lines of a system file

head -n 5 /etc/passwd

View the last 5 lines of a system file

tail -n 5 /etc/passwd

5.5. Obtaining Command Assistance & Documentation: Help Systems.

Learn how to independently research command parameters and system utility usage.

View single-line summary description of a command

whatis cp

whatis ls

View short help summary option for built-in/gnu utilities

cp --help

Open full manual page for a command (Press 'q' to exit manual view)

man mkdir

Search manual database for a keyword

man -k "directory"

6.6. Directory & File Removal: Safe Cleanup.

Safely delete test files and empty directories using interactive confirmation flags.

Navigate back to home directory

cd ~

Interactively remove files inside data/

rm -i ~/cli_lab/data/day1_backup.txt ~/cli_lab/data/dataset1.csv

Remove empty directories

rmdir ~/cli_lab/data

Recursively remove the remaining lab directory and contents with confirmation

rm -r -i ~/cli_lab

Command Quick-Reference Summary

Category	**Command**	**Primary Function**	**Frequently Used Options**
Navigation	pwd	Print working directory	*(None standard)*
	ls	List directory contents	-l (long format), -a (show hidden), -h (human-readable sizes)
	cd	Change directory	~ (home), .. (parent folder), - (previous directory)
Management	mkdir	Make directory	-p (create parent paths as needed), -v (verbose)

Category	Command	Primary Function	Frequently Used Options
	touch	Create empty file / update timestamp	*(None standard for simple creation)*
	cp	Copy files/directories	-r (recursive for folders), -i (interactive mode)
	mv	Move or rename files/directories	-i (interactive mode), -v (verbose)
	rm	Remove files or directories	-r (recursive), -f (force), -i (interactive prompt)
Viewing	cat	Display entire file	-n (number all lines)
	less	Interactive paginated viewer	-N (show line numbers); press / to search, q to quit
	head / tail	View top or bottom lines	-n <count> (number of lines to output)
Help System	man	Comprehensive manual page	Section numbers (e.g., man 1 ls, man 2 read)
	whatis	One-line summary lookup	*(None standard)*

Category	Command	Primary Function	Frequently Used Options
	--help	Inline flag for quick options	Added after almost any command (e.g., rm --help)